

## MIT 18.06 - 2.8

### Problem 21.1

- a)  $B$  has rank 2 because it has two distinct non zero eigenvalues
- b)  $\det(B) = 0$  because it has eigenvalue 0, therefore  $\det(B^T B) = 0$
- c) Cant be found from given information
- d)  $Bx = \lambda x$  so  $B^2 x = B(Bx) = B\lambda x = \lambda Bx = \lambda^2 x$   
 $(B^2 + I)x = B^2 x + Ix = B^2 x + x = \lambda^2 x + x = (\lambda^2 + 1)x$   
 $\therefore$  eigenvalue of  $B^2 + I$  is  $\lambda^2 + 1$  where  $\lambda$  are the eigenvalues of  $B$ .

$$\begin{aligned}(B^2 + I)x &= (\lambda^2 + 1)x \text{ then } (B^2 + I)^{-1}(B^2 + I)x = x \\ &= (B^2 + I)^{-1}(\lambda^2 + 1)x = x \\ &= \frac{(B^2 + I)^{-1}}{(\lambda^2 + 1)} x\end{aligned}$$

So the eigenvalues of  $(B^2 + I)^{-1}$  are:-

$$\frac{1}{(0^2 + 1)} = 1, \quad \frac{1}{1^2 + 1} = \frac{1}{2}, \quad \text{and} \quad \frac{1}{2^2 + 1} = \frac{1}{5}$$

### Problem 21.2

Eigenvalues of A are 1, 4 and 6.

Eigenvalues of B are the solution to:-

$$\det \begin{pmatrix} -\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 3 & 0 & -\lambda \end{pmatrix} = 0$$

$$\begin{aligned} & -\lambda((2-\lambda)(-\lambda)) + (-1)(2-\lambda)(3) \\ & -\lambda(-2\lambda + \lambda^2) + (-6 + 3\lambda) \\ & 2\lambda^2 - \lambda^3 - 6 + 3\lambda \end{aligned}$$

$$(-\lambda)(2-\lambda)(-\lambda) - (3)(2-\lambda) = 0$$

$$\lambda^2(2-\lambda) - (3)(2-\lambda) = 0$$

$$(2-\lambda)(\lambda^2 - 3) = 0$$

$$(2-\lambda)(\lambda + \sqrt{3})(\lambda - \sqrt{3}) = 0$$

$$\lambda = 2$$

$$\lambda = -\sqrt{3}$$

$$\lambda = \sqrt{3}$$

Eigenvalues of C are the solutions to:-

$$\det \begin{pmatrix} 2-\lambda & 2 & 2 \\ 2 & 2-\lambda & 2 \\ 2 & 2 & 2-\lambda \end{pmatrix} = 0$$

$$\begin{aligned} & (2-\lambda)((2-\lambda)^2 - 4) + 2(-1(2(2-\lambda) - 4)) \\ & + 2(4 - 2(2-\lambda)) = 0 \end{aligned}$$

$$\begin{aligned} & (2-\lambda)^3 - 4(2-\lambda) + 2(-4 + 2\lambda + 4) \\ & + 2(4 - 4 + 2\lambda) = 0 \end{aligned}$$

$$(4 - 4\lambda + \lambda^2)(2 - \lambda) - 8 + 4\lambda + 4\lambda + 4\lambda = 0$$

$$(8 - 4\lambda - 8\lambda + 4\lambda^2 + 2\lambda^2 - \lambda^3) - 8 + 4\lambda + 12\lambda = 0$$

$$-\lambda^3 + 6\lambda^2 = 0$$

$$\lambda^2(-\lambda + 6) = 0$$

$$\lambda = 0$$

$$\lambda = 0$$

$$\lambda = 6$$